10

The Human Side

Living and Thinking in a Quantum Universe

In the course of a long life, each of us will occasionally encounter a difficult decision we must make. Stay single or get married? Go for a run or have another doughnut? Go to grad school or enter the real world?

Wouldn't it be nice to be able to choose both sides, rather than picking one? Quantum mechanics suggests a strategy: whenever you have a decision to make, you can do so by consulting a quantum random-number generator. Indeed, there is an app available for iPhones called Universe Splitter that can be used for this very purpose. (As Dave Barry says, I swear I am not making this up.)

Let's say you have a choice to make: "Should I get pepperoni or sausage on my pizza?" (And let's say you have too much restraint to give the obvious answer of asking for both on the same pizza.) You can fire up Universe Splitter, where you will see two text boxes, into which you can type "pepperoni" and "sausage." Then hit the button, and your phone will send a signal through the internet to a laboratory in Switzerland, where a photon is sent toward a beam splitter (essentially a partially silvered mirror that reflects some photons and lets others through). According to the Schrödinger equation, the beam splitter turns the photon's wave function into two components going left and right, each of which heads toward a different detector. When either detector notices a photon, it produces a readout that becomes entangled with the environment, quickly leading to decoherence and branching the wave function in two. The copy of you in the branch where the photon went left sees their phone flash with the